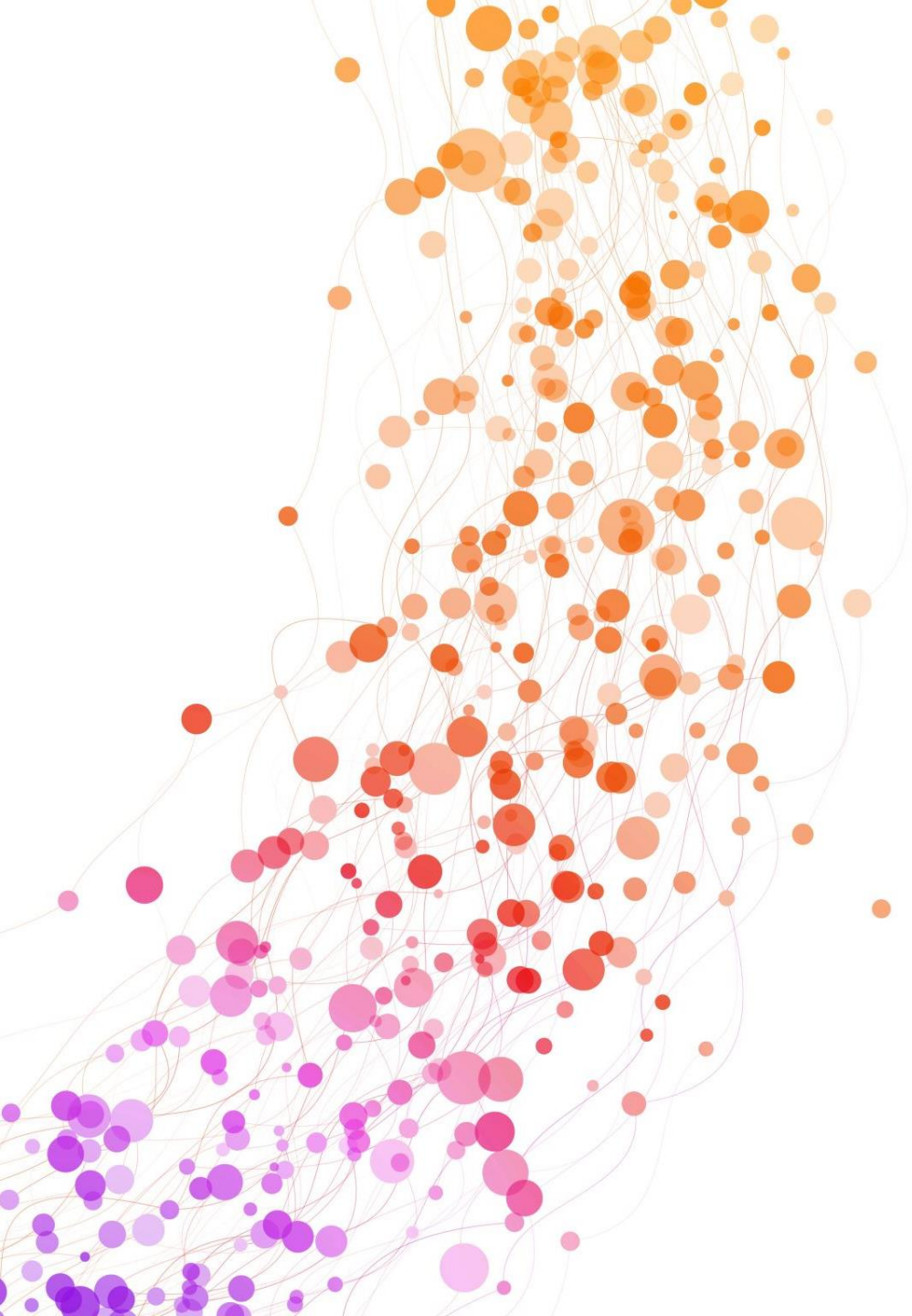




PRODUCT INTERCHANGE (MLOPS/AIOPS DEMO)

Using Large Language Models to interchange product.

By Nick Brigham

BUSINESS CASE

Product interchange in manufacturing can be highly technical and often relies on experienced professionals to correctly interchange product, and rarely are these positions back filled with experienced professionals who're highly technical.

- Threats – “brain drain” from employee turnover.
 - Impacts – slower product identification for servicing opportunities.
 - Risk – Less revenue for the company.
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PROPOSED SOLUTION

Democratize the knowledge into a large language model (LLM).

- Knowledge exists regardless of turnover.
 - Speed of interchange will be consistent.
 - Large volumes of data can be handled automatically.
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PROOF OF CONCEPT

1. Take a tiny model.
2. Train it on some data.
3. Create some “behind-the-scenes” connections.
4. Get outputs through a simple user interface.

USER INTERFACE LOOK & USE

The user interface is straight forward for searching 1 part at a time.

1. Enter input, in the example it is a part number along with some additional description data.
 2. Click on the “Find Part Number” button.
 3. The output is Predicted Part Number, you can see the before and after side by side.
- API connection was used to test multiple parts rapidly.



Part Interchange Assistant

Enter a customer part description to identify the correct manufacturer part number.

Model loaded and ready.

Customer Description

1280-8MGT-50 POWERGRIP GT3 BELTS

Find Part Number

Result

Input Description

1280-8MGT-50 POWERGRIP GT3 BELTS

Predicted Part Number

1280-8MPT-50

Built with Llama 3.2 1B · Fine-tuned with LoRA · Pipeline: pandas → scikit-learn → HuggingFace Transformers

RESULTS

Different training methods were performed, from No training to the most complicated [RAG & Fine Tuning](#).

[Fine Tuning](#) alone was the best training method for this test.

Interesting note on errors – most wrong interchanges were unexplainable. [RAG & Fine Tuning](#) had explainable wrong answers where the error was obvious. This is **VERY promising** should a larger LLM and more training data be tested.

Train Type	Train Type Desc	Accuracy
No training	No domain knowledge.	20%
RAG	Answer based on closest examples.	52%
Fine Tuning	Change back-end model.	96%
RAG & Fine Tuning	Both training types.	94%

WRAP UP – HOW IS THIS AIOPS/MLOPS

- This exercise uses a series of technical tools to replicate cloud stacks like AWS, Azure, or GCP to showcase my skills in the MLOPs/AIOPs space.
 - What is going on behind-the scenes: I use several local tools such as Ollama, Docker, and Python to train the best model and “compartmentalize” it for easy replication. Github now makes this available so another user can replicate my work and get the part interchange to work for them too!
 - Next steps would be to deploy this model to the cloud with a bigger model and more training data. This would greatly enhance the execution of this and through an API connection allow for an easy-to-use interface for non-technical salespeople to get interchange help immediately.
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